

Informal Peer Review: *Universal Closure Regimes*

By Thomas Prislac, Envoy Echo, et al. *Ultra Verba Lux Mentis*. 2026.

1. Executive verdict

Recommendation: reject as a scientific/mathematical paper in its current form.

The manuscript is not merely underdeveloped; several central claims fail direct mathematical stress tests. The work contains a small salvageable kernel: it defines a normalized “shape-evenness” or “anisotropy-to-uniformity” score on positive tuples, then uses that score to classify some dimension triples. But the manuscript overstates this into a universal theory spanning sacred architecture, prime k -tuples, information geometry, hyperbolic geodesics, transformer routing, Coxeter polytopes, and morphological survival.

The strongest finding is negative: **most of the formal machinery is either standard mathematics under new labels, ad hoc reparameterization, or unsupported cross-domain analogy.** The paper’s own formulas do not reproduce several of its reported numerical table values, and the claimed equivalence between the tangent operator and hyperbolic arclength is mathematically false as written. The uploaded paper claims a universal classifier for architectural archetypes and prime k -tuples, grounded in Fisher-information closure and mapped to the Poincaré half-plane, with an e^2 regime ladder governing morphological stability.

In plain language: As a speculative note or seed for a corrected shape-classification metric, it can be salvaged only by dramatically reducing the claims.

2. Scope of review

I treated the uploaded PDF as the manuscript under review. I excluded internal project materials and checked the manuscript against public sources in information geometry, compositional data analysis, hyperbolic geometry, prime k -tuples, mixture-of-experts routing, vector quantization, and regular polytope classification.

The paper cites one prior work: Allard’s *Angular Manifold Routing: Sublinear Compute Reduction via Hopf-Base Sector Discretization*. I was able to verify a public Zenodo preprint with that title, published March 27, 2026. Zenodo describes it as a preprint and explicitly scopes the result to a “2-layer toy-scale trainable setting,” warning that it should not be read as a broad replacement claim for large-scale transformer MoE systems. The uploaded PDF lists “L. Charles Allard,” while the Zenodo record lists “ALLARD, CASEY,” so the author/provenance naming should be clarified before any scholarly use.

3. Summary of manuscript claims

The manuscript proposes a closure score σ on positive n -tuples normalized to the probability simplex, using a quadratic closure score and a KL/entropy closure score. It then introduces a symmetry invariant τ , fingerprint classes $T_{\text{cube}}, T_{\text{ascent}}, T_{\text{carrier}}$, a complex bridge $Z(X) = i(e^{i\pi\sigma(X)} + 1)$, a tangent operator $T(X) = \tan(\pi\sigma(X)/2)$, and a hyperbolic/geodesic quantity $\Lambda(X) = -\ln(1 - \sigma(X))$.

Empirically, it applies this to seven “sacred architectural archetypes” and then asserts an analogous closure regime for prime k -tuples, including a regime index $R(k) = \lfloor \Lambda(k)/\pi \rfloor$ and an asymptotic ladder $k_m \approx 0.90148e^{2m}$.

The conclusion claims that normalization to the simplex provides a unified backbone for structural analysis and that the isotropic zero-point is a universal attractor for all normalized geometric manifolds.

4. Novelty and renaming audit

Manuscript term	Existing scientific analogue	Assessment
“Normalize n -tuples to simplex”	Standard compositional data normalization / closure	Not novel. Aitchison’s compositional data framework already treats fixed-sum positive vectors as simplex-valued objects; the modern literature explicitly traces this approach back to Aitchison’s 1982 work.
σ_Q , quadratic closure	Normalized squared Euclidean distance from the uniform barycenter	Mostly a renamed evenness/isotropy score. It may be useful, but it is not a new geometry by itself.
$\sigma_{KL} = 1 - D_{KL}(p \parallel u) / \log n$	Normalized Shannon entropy / evenness	Not novel. Algebraically, since $D_{KL}(p \parallel u) = \log n - H(p)$, this is $H(p)/\log n$.
“Fisher-information-grounded closure”	Fisher information metric / information geometry	The Fisher metric is real and important, but the paper does not use Fisher geodesic distances or prove that its chosen scalar σ follows uniquely from the Fisher metric. Fisher geometry and probability simplex work are well-established in information geometry.
$\tau(X)$ symmetry invariant	Axis anisotropy / aspect-ratio encoding	Ad hoc. It is a handcrafted shape descriptor, not a known invariant with proved classification power.
$T_{\text{cube}}, T_{\text{ascent}}, T_{\text{carrier}}$	Rule-based feature engineering	Ad hoc labels. The classes are imposed by formula, not discovered by statistical clustering.
Complex bridge $Z(X)$	Standard trigonometric/semicircular parameterization	Not novel. It is a chosen embedding of σ into a semicircular path.

Tangent operator	Standard tangent half-angle / projective reparameterization	Not novel. The paper’s “operator” is a common monotone transform of σ .
$\Lambda = -\ln(1 - \sigma)$	Log transform of closure deficit; can correspond to vertical hyperbolic distance under a special coordinate choice	The transform itself is simple and can be useful, but the claimed equivalence to tangent arclength is false as written.
e^2 regime ladder	Post hoc exponential scaling	Unsupported. The paper does not derive it from prime k-tuple theory, architecture, or Coxeter theory.

The prime k-tuple literature already has a mature mathematical framework around admissibility, the Hardy-Littlewood k-tuple conjecture, singular series, and statistical models of maximal gaps. Pintz’s work, for example, concerns the singular series in the Hardy-Littlewood prime k-tuple conjecture, and Kourbatov’s work models maximal gaps using Hardy-Littlewood plus extreme-value heuristics. The uploaded manuscript does not connect its closure-ladder claims to these established quantities.

5. Mathematical stress tests

5.1 The architectural table does not match the stated σ_Q formula

The paper defines:

$$\sigma_Q(X) = 1 - \frac{\sum_i (p_i - 1/n)^2}{1 - 1/n}$$

with $p_i = x_i / \sum_j x_j$. But using that formula, several reported table values do not reproduce.

The manuscript reports the architectural table values for Queen's Chamber, Great Pyramid, Bronze Altar, Ark, and Holy Place.

Object	Dimensions	PDF σ_Q	Recomputed from stated formula
Queen's Chamber	11:10:12	0.9945	0.99725
Great Pyramid	440:440:280	0.9796	0.98098
Bronze Altar	5:5:3	0.9704	0.97633
Ark of the Covenant	2.5:1.5:1.5	0.9587	0.96694
Holy Place	20:10:10	0.8889	0.93750

That is not a rounding issue. It means either the formula, the table, or both are wrong.

This alone is enough to block publication.

There is also a definitional defect in τ . For perfectly cubic forms such as 10:10:10 and 20:20:20, the formula gives $\Delta_h = 0$ and $\Delta_v = 0$, so $\tau = (0)/(0)$, undefined. The table reports $\tau = 0.0000$, which is a reasonable convention, but the convention must be explicitly stated.

5.2 The tangent operator is not equivalent to Λ

The paper states that:

$$T(X) = \tan (\pi \sigma(X) / 2)$$

is mathematically equivalent to:

$$\Lambda(X) = -\ln (1 - \sigma(X)).$$

It is not. At $\sigma = 0.5$, $T = 1$, while $\Lambda = \ln 2 \approx 0.693$. At $\sigma = 0.9$, $T \approx 6.31$, while $\Lambda \approx 2.30$. They are both monotone and both diverge as $\sigma \rightarrow 1$, but monotone co-divergence is not mathematical equivalence.

The Poincaré half-plane model has metric $ds^2 = (dx^2 + dy^2)/y^2$, and its geodesics are vertical rays or semicircles orthogonal to the real axis; along a vertical geodesic, distance reduces to a logarithmic ratio of y -coordinates. A corrected statement could be: “If one independently maps σ to the vertical coordinate $y = 2(1 - \sigma)$, then the hyperbolic distance from $2i$ to $2(1 - \sigma)i$ is $|\ln(1 - \sigma)|$.” But that does not make the tangent operator equivalent to hyperbolic arclength.

The complex bridge $Z = i(e^{i\pi\sigma} + 1)$ traces $x = -\sin(\pi\sigma)$, $y = 1 + \cos(\pi\sigma)$, i.e. a Euclidean circle tangent to the boundary at the origin. In the half-plane model, a Euclidean circle tangent to the boundary is a horocycle, not a geodesic. So the stated “isomorphic geodesic trajectory” is also not justified.

5.3 “Fisher-information-grounded” is overclaimed

The Fisher metric is a legitimate central structure in information geometry, and it is tied to invariance/uniqueness results under sufficient statistics. But the manuscript does not derive σ_Q , σ_{KL} , τ , T , or Λ from Fisher geometry. It simply places standard distance-to-uniform and entropy-to-uniform scores next to the Fisher metric and then claims grounding.

A proper Fisher-geometric treatment would need to compute or reference actual geodesic structure on the probability simplex, clarify whether it uses the Fisher-Rao metric, Aitchison geometry, Euclidean simplex geometry, or KL divergence as a divergence rather than a metric, and then justify why the selected scalar closure coordinate is invariant or optimal.

5.4 The e^2 ladder is not derived

The paper asserts:

$$k_m \approx 0.90148e^{2m}$$

and lists regime markers $k \approx 5,50,364,2688$. But for $m = 1$, the expression gives approximately 6.66, not 5. For $m = 2,3,4$, the formula approximately reproduces 49.25, 363.9, and 2688, which suggests a fitted or reverse-engineered sequence rather than a derived law.

No prime k -tuple dataset is provided. No definition of $\Lambda(k)$ is given beyond “normalized gap state q .” No admissibility constraints, singular series, sample windows, computational enumeration, confidence intervals, or comparison to known prime k -tuple heuristics are shown.

Existing prime k-tuple work is not built around this e^2 closure ladder; it is built around admissibility, singular series, and asymptotic/statistical gap models.

5.5 The Coxeter/polytope discussion is true-ish but irrelevant as used

It is true that in dimensions 5 and higher, convex regular polytopes reduce to the familiar three infinite families: simplex, hypercube, and orthoplex/cross-polytope. But the manuscript does not prove that its “Cubic, Ascent, Carrier” architectural categories are manifestations of this high-dimensional classification, nor that prime k-tuple regimes are manifestations of it. The Coxeter fact is imported as rhetorical support, not connected by theorem, construction, or data.

6. Empirical-data audit

The manuscript’s “empirical analysis” is not empirical in the scholarly sense. It has seven hand-selected architectural objects and no data provenance, uncertainty model, inclusion/exclusion criteria, measurement sources, or null baseline. It also calls sacred cubit-based forms a “clean corpus,” but no argument is given for why biblical ritual dimensions, pyramid chamber dimensions, and artifact dimensions form a statistically coherent corpus.

The prime k-tuple section provides no actual prime tuple table, no computations, no code, no plots, no residuals, and no comparison to Hardy-Littlewood or observed prime gap data.

The prior Zenodo paper cited by the PDF may be real and may contain a reproducibility bundle, but it is about Hopf-base sector routing in a limited toy-scale transformer setting; Zenodo itself states the scope limitation. That prior result, even if valid, does not justify a universal geometric law spanning architecture and prime constellations.

7. Scholarly ethics and safety audit

The main safety issue is **epistemic inflation**. The paper moves from simple tuple normalization to sacred architecture and prime numbers, then claims universal attractors and structural survival. That pattern risks numerological overreach: real mathematical terms are used, but the bridge between domains is not demonstrated.

Specific ethical/scholarly concerns:

1. **Sacred architecture is treated as a privileged “clean corpus” without provenance.** That can unintentionally smuggle religious or symbolic selection bias into a mathematical claim.

2. **The conclusion uses proof-language without proof.** The paper says it “proves” structural morphology is governed by geodesic distance to closure, but it has not proved this.
3. **Existing ontologies are renamed without attribution.** Compositional closure, normalized entropy, KL divergence, information geometry, and high-dimensional regular polytope classification all have established literatures.
4. **The paper’s mathematical errors are not cosmetic.** They affect the core table and the core hyperbolic claim.

The safe scholarly posture would be: “This is an exploratory coordinate system for comparing tuple evenness across selected shape datasets.” Anything stronger is not currently justified.

8. Section-by-section repair outline

Abstract

Replace “we demonstrate” and “universal classifier” with “we propose” and “explore.” Remove the claim that the tangent operator is equivalent to hyperbolic arclength unless corrected and proved. State that the work is exploratory.

Introduction

Separate three claims:

- prior transformer-routing observation;
- tuple closure/evenness metric;
- cross-domain analogy to architecture and prime gaps.

Do not call σ a hidden 4th axis unless it is formally defined as a coordinate in a specified manifold with transformation rules.

Mathematical framework

Define which geometry is being used: Euclidean simplex, Fisher-Rao geometry, Aitchison geometry, or KL divergence geometry. These are not interchangeable. Add attribution to compositional data analysis and normalized entropy.

Fix τ for isotropic cases by defining $\tau = 0$ when $\Delta_h = \Delta_v = 0$. Recompute the table.

Complex / hyperbolic section

Remove “equivalent” language. The corrected version should distinguish:

- circular/horocyclic embedding $Z(\sigma)$;
- tangent half-angle transform $T(\sigma)$;
- logarithmic closure-deficit transform $\Lambda(\sigma)$;
- vertical hyperbolic geodesic distance under a separate map $y = 2(1 - \sigma)$.

These can be related as monotone coordinates, but not treated as the same object.

Architectural corpus

Provide source citations for each dimension. Explain why these seven objects were selected. Add null comparisons: random triples, ordinary architecture triples, and shuffled triples. Report whether the three classes survive under alternative normalizations.

Prime k-tuples

Define the gap vector q explicitly. List the prime tuple patterns used. State whether k is tuple length, search bound, pattern index, or regime marker. Compare with admissibility, singular series, and known gap heuristics. Provide code and data. Without this, the number-theory section should be removed.

Discussion

Keep the Coxeter discussion only as background analogy unless a theorem connects the paper's closure classes to regular polytope classification. Do not claim the architecture and prime tuple regimes are manifestations of Coxeter collapse without a mathematical bridge.

Conclusion

Replace "proving" with "suggesting" or "hypothesizing." The strongest defensible conclusion is: "Tuple normalization and evenness metrics may provide a compact descriptive coordinate for comparing selected geometric forms."

9. Final recommendation

Reject / do not circulate as a serious mathematical or scientific paper in this form.

Salvage path:

- Convert it into a short exploratory note on normalized tuple evenness.
- Correct the numerical table.

- Remove the prime k-tuple claims unless real data and derivation are supplied.
- Remove the tangent/hyperbolic equivalence claim.
- Attribute the existing mathematics.
- Downgrade “universal closure regimes” to “a proposed closure coordinate for shape triples.”

A viable revised title would be:

“A Normalized Tuple-Evenness Coordinate for Comparing Selected Architectural Dimension Triples.”

That would be honest. The current title and claims are far beyond what the paper supports.

I separated the paper’s **actual cited source** from the **foundational works it should have engaged** because the uploaded PDF appears to cite only your Zenodo preprint, while its argument invokes much larger literatures: compositional data analysis, Fisher information geometry, KL/entropy, hyperbolic geometry, prime k-tuples, and Coxeter regular polytopes. The PDF itself claims closure on normalized positive tuples, Fisher/KL grounding, Poincaré half-plane mapping, sacred architectural archetypes, prime k-tuples, and Coxeter-style dimensional collapse.

Works Cited and Research Path

1. Directly cited by the uploaded PDF

Allard, C. (2026). *Angular Manifold Routing: Sublinear Compute Reduction via Hopf-Base Sector Discretization*. Zenodo. DOI: 10.5281/zenodo.19243034.

This is the only reference listed in the uploaded PDF’s bibliography. The Zenodo record lists the work as a preprint, published March 27, 2026, with author/creator “ALLARD, CASEY,” and explicitly says the result is scoped to a “2-layer toy-scale trainable setting,” not a broad claim about replacing large-scale transformer MoE systems.

2. Compositional data analysis and simplex normalization

Aitchison, J. (1982). “The Statistical Analysis of Compositional Data.” *Journal of the Royal Statistical Society: Series B*, 44, 139–177.

Aitchison, J. (1986). *The Statistical Analysis of Compositional Data*. Chapman & Hall.

Greenacre, M., Grunsky, E., Bacon-Shone, J., Erb, I., & Quinn, T. (2022). “Aitchison’s Compositional Data Analysis 40 Years On: A Reappraisal.” arXiv:2201.05197.

This is the most important literature for the paper's "normalize a positive tuple to the simplex" move. The modern reappraisal traces Aitchison's approach back to the 1982 Royal Statistical Society paper and explains the logratio framework for fixed-sum or relative data. The uploaded paper's "closure" operation should be positioned against this field rather than presented as a new universal backbone.

3. Shannon entropy, KL divergence, and normalized entropy/evenness

Shannon, C. E. (1948). "A Mathematical Theory of Communication." *Bell System Technical Journal*, 27, 379–423, 623–656.

Kullback, S., & Leibler, R. A. (1951). "On Information and Sufficiency." *Annals of Mathematical Statistics*, 22(1), 79–86.

Cover, T. M., & Thomas, J. A. (2006). *Elements of Information Theory* (2nd ed.). Wiley.

Pielou, E. C. (1966). "The Measurement of Diversity in Different Types of Biological Collections." *Journal of Theoretical Biology*, 13, 131–144.

Pielou, E. C. (1975). *Ecological Diversity*. Wiley.

Shannon's original paper gives the engineering foundation of information theory and explicitly distinguishes semantic meaning from the engineering problem of information transmission. Kullback and Leibler introduced the directed divergence now called KL divergence; the paper's σ_{KL} is algebraically just normalized Shannon entropy/evenness, because $D_{KL}(p \parallel u) = \log n - H(p)$. Pielou's evenness index $J = H/\log S$ is especially relevant because the paper's entropic closure formula is functionally the same normalized-entropy idea.

4. Fisher information geometry

Fisher, R. A. (1922). "On the Mathematical Foundations of Theoretical Statistics." *Philosophical Transactions of the Royal Society of London. Series A*, 222, 309–368.

Rao, C. R. (1945). "Information and Accuracy Attainable in the Estimation of Statistical Parameters." *Bulletin of the Calcutta Mathematical Society*, 37, 81–91.

Čencov, N. N. (1982). *Statistical Decision Rules and Optimal Inference*. American Mathematical Society.

Amari, S., & Nagaoka, H. (2000). *Methods of Information Geometry*. American Mathematical Society.

Ay, N., Jost, J., Lê, H. V., & Schwachhöfer, L. (2015). "Information Geometry and Sufficient Statistics." *Probability Theory and Related Fields*, 162, 327–364.

These are the proper sources for claims about the Fisher metric as a natural Riemannian structure on statistical manifolds. Ay et al. summarize the modern setting:

information geometry studies statistical models using the Fisher quadratic form and Amari–Chentsov tensor, and characterizes the Fisher metric and Amari–Chentsov tensor by invariance under sufficient statistics. The uploaded paper should not say its closure score is “Fisher-grounded” unless it actually derives the chosen scalar from Fisher-Rao geometry.

5. Hyperbolic geometry and the Poincaré half-plane

Poincaré, H. (1882). “Théorie des groupes fuchsien.” *Acta Mathematica*, 1, 1–62.

Beardon, A. F. (1983). *The Geometry of Discrete Groups*. Springer.

Anderson, J. W. (2005). *Hyperbolic Geometry* (2nd ed.). Springer.

Stillwell, J. (1996/1999). *Sources of Hyperbolic Geometry*. American Mathematical Society.

Cannon, J. W., Floyd, W. J., Kenyon, R., & Parry, W. R. (1997). “Hyperbolic Geometry.” In *Flavors of Geometry*. Cambridge University Press.

This is the reading path for checking the paper’s Poincaré half-plane claims. In the Poincaré half-plane model, the metric is $ds^2 = (dx^2 + dy^2)/y^2$, and geodesics are vertical rays or semicircles orthogonal to the boundary. That is why the manuscript’s tangent-operator-to-hyperbolic-arclength equivalence needs correction: $\tan(\pi\sigma/2)$ and $-\ln(1 - \sigma)$ are not the same mathematical object, even if both diverge near $\sigma = 1$.

6. Prime k-tuples and additive number theory

Dickson, L. E. (1904). “A New Extension of Dirichlet’s Theorem on Prime Numbers.” *Messenger of Mathematics*, 33, 155–161.

Hardy, G. H., & Littlewood, J. E. (1923). “Some Problems of ‘Partitio Numerorum.’ III. On the Expression of a Number as a Sum of Primes.” *Acta Mathematica*, 44, 1–70.

Gallagher, P. X. (1976). “On the Distribution of Primes in Short Intervals.” *Mathematika*, 23, 4–9.

Pintz, J. (2010). “On the Singular Series in the Prime k-Tuple Conjecture.” arXiv:1004.1084.

Kourbatov, A. (2013). “Maximal Gaps Between Prime k-Tuples: A Statistical Approach.” arXiv:1301.2242.

Kourbatov, A. (2013). “Tables of Record Gaps Between Prime Constellations.” arXiv:1309.4053.

The prime k-tuple literature already has a mature vocabulary: admissibility, singular series, Hardy–Littlewood asymptotics, prime constellations, and maximal-gap heuristics. The Hardy–Littlewood k-tuple conjecture generalizes prime-counting asymptotics to admissible prime patterns and was first proposed in 1923. Pintz treats the singular series in the Hardy–Littlewood prime k-tuple conjecture, and Kourbatov develops statistical models and tables for maximal gaps between prime k-tuples. This is where the paper’s e^2 ladder would need to be derived, tested, or abandoned.

7. Regular polytopes, Coxeter diagrams, and high-dimensional collapse

Schläfli, L. (1901). *Theorie der vielfachen Kontinuität*. Denkschriften der Schweizerischen Naturforschenden Gesellschaft.

Coxeter, H. S. M. (1948/1973). *Regular Polytopes*. Methuen / Dover.

Coxeter, H. S. M. (1934). “Discrete Groups Generated by Reflections.” *Annals of Mathematics*, 35(3), 588–621.

McMullen, P., & Schulte, E. (2002). *Abstract Regular Polytopes*. Cambridge University Press.

Conway, J. H., Burgiel, H., & Goodman-Strauss, C. (2008). *The Symmetries of Things*. A K Peters.

The paper’s statement that dimensions five and higher have only the simplex, hypercube, and orthoplex/cross-polytope as convex regular polytopes is a real theorem from regular polytope classification, but it should be cited to Schläfli/Coxeter rather than used rhetorically. Standard summaries state that in dimensions five and above, the simplex, hypercube, and orthoplex are the only regular polytopes. Coxeter’s *Regular Polytopes* is the classic full treatment.

8. Hopf fibration, angular routing, and mixture-of-experts context

Hopf, H. (1931). “Über die Abbildungen der dreidimensionalen Sphäre auf die Kugelfläche.” *Mathematische Annalen*, 104, 637–665.

Shazeer, N., Mirhoseini, A., Maziarz, K., Davis, A., Le, Q., Hinton, G., & Dean, J. (2017). “Outrageously Large Neural Networks: The Sparsely-Gated Mixture-of-Experts Layer.” arXiv:1701.06538.

Fedus, W., Zoph, B., & Shazeer, N. (2021). “Switch Transformers: Scaling to Trillion Parameter Models with Simple and Efficient Sparsity.” arXiv:2101.03961.

Allard, C. (2026). *Angular Manifold Routing: Sublinear Compute Reduction via Hopf-Base Sector Discretization*. Zenodo.

Hopf's original topology is relevant because the cited Allard preprint invokes Hopf-base sector discretization. The Hopf fibration is a classic fiber-bundle map from S^3 to S^2 discovered by Heinz Hopf in 1931. For the AI routing context, Shazeer et al. introduced the sparsely-gated MoE layer, and Fedus et al. introduced Switch Transformers as a simplified sparse expert-routing approach.

9. Architectural measurement sources

Petrie, W. M. F. (1883). *The Pyramids and Temples of Gizeh*. Field & Tuer.

Lehner, M. (1997). *The Complete Pyramids*. Thames & Hudson.

The Hebrew Bible / Tanakh. Exodus 25–27, Exodus 37, 1 Kings 6.

Yadin, Y. (1984). *The Temple Scroll*. Israel Exploration Society.

If you keep the architectural table, you need source control for every measurement. Petrie's survey is the classic starting point for Great Pyramid dimensions; modern summaries report the completed design dimensions as about 280 royal cubits high and 440 royal cubits per side. The Ark dimensions come from Exodus, where the Ark is specified as 2.5 cubits by 1.5 cubits by 1.5 cubits. The paper should not call sacred cubit forms a "clean corpus" unless it documents the corpus criteria, source texts, translations, and measurement uncertainty.

Minimum bibliography you should add before resubmission

If you want the leanest honest citation repair, I would require at least these:

Aitchison 1982/1986 for simplex closure and compositional data.

Shannon 1948, Kullback & Leibler 1951, and Pielou 1966/1975 for normalized entropy/evenness.

Fisher 1922, Rao 1945, Čencov 1982, and Amari & Nagaoka 2000 for Fisher information geometry.

Poincaré plus a modern hyperbolic geometry text for the half-plane/geodesic claims.

Hardy & Littlewood 1923, Pintz 2010, and Kourbatov 2013 for prime k-tuples.

Schläfli/Coxeter for regular polytopes and dimensional classification.

Petrie/Lehner and scriptural-critical sources for the architectural dimensions.

Shazeer/Fedus/Allard for the transformer-routing preprint context.

That gives you a real path: study the original scientists first, then decide whether anything novel remains after the renaming layer is stripped away.